# 1. Clinical Question & Core Hypothesis

## Clinical Question

The Ontologist highlights frequent discordance among noninvasive readouts (urine tumor DNA methylation), systemic dissemination measures (CTCs), and local immune organization (Immunoscore/TLS). Why do some bladder tumors show dense local immune architecture yet low shedding into urine/blood and still experience early progression? 2) Tier 1 paths used: I built the hypothesis on the three Tier-1 axes: urine DNA methylation biomarkers (noninvasive tumor shedding), circulating tumor cells (CTCs, systemic dissemination/readout), and immune contexture (operationalized here as TLS-like lymphoid aggregates). These represent complementary spatial scales (luminal shedding, bloodstream dissemination, and tissue immune microenvironment) most directly relevant to surveillance and progression. 3) Tier 2 support: Mechanistic anchors from Tier 2 (MSH6, TGF-β signaling, SDC2/TFPI2 methylation, HDL-C) plausibly modulate both shedding and immune organization — e.g., MSH6-driven mutational load may alter neoantigenicity; TGF-β can drive immune exclusion and EMT that changes shedding; locus-specific methylation panels (SDC2/TFPI2) inform assay sensitivity; HDL-C reflects systemic metabolic/inflammatory state. These support testable mechanistic modifiers but are kept out of the core hypothesis to preserve focus. 4) Endpoint choice (PFI): I chose progression-free interval because urine methylation and CTCs are primarily surveillance/monitoring tools whose clinical utility is evaluated by detecting progression/recurrence earlier than imaging or clinical symptoms; PFI best captures the predictive value of discordant shedding/immunity patterns. 5) Deliberate exclusions: I excluded broader implementation/AI considerations, stool/PSA cross-cancer markers, and other exploratory modalities (Tier 3) to keep the hypothesis sharply mechanistic and testable. I also excluded naming specific signaling mediators (e.g., TGFB1) in the core claim to satisfy the constraint of max two anchor molecular features, relegating them to secondary_modifiers for follow-up testing?

## Working Hypothesis

Among bladder cancer patients, TLS-like lymphoid aggregates (tissue immune architecture) and noninvasive shedding readouts (urine DNA methylation biomarkers and circulating tumor cells, CTCs) are two anchor axes that together define a discordant phenotype with distinct prognostic implications. Specifically, tumors that are TLS-positive but exhibit low urine methylation signal and low CTC counts (TLS+/shedding-low) will have a shorter progression-free interval (PFI) than TLS-positive tumors with higher shedding/CTC levels (TLS+/shedding-high). This paradox arises because TLS-positive tumors can harbor a spatially compartmentalized immune microenvironment in which an immune-excluded tumor core enables occult invasive clones and EMT/TGF-β–driven progression, while local shedding into urine or bloodstream remains suppressed. The core hypothesis remains anchored on the two primary surveillance axes (urine DNA methylation biomarkers and CTCs) and posits that discordance with TLS density predicts earlier progression, independent of standard clinicopathologic stage. Secondary modifiers (MSH6-related mutational/neoantigen landscape, TGF-β–driven immune exclusion, SDC2/TFPI2 methylation panels affecting assay sensitivity, and systemic HDL-C) may influence the strength of the TLS–shedding–progression relationship but are not primary drivers of the core effect. The working model implies that in TLS+/shedding-low tumors, the immune milieu is reorganized in a way that suppresses local shedding yet permits or accompanies invasive subclones, leading to accelerated progression despite apparent local immune organization.

| Element | Value |
|---|---|
| Imaging feature | TLS-like lymphoid aggregates |
| Anchor gene(s) | urine DNA methylation biomarkers, circulating tumor cells (CTCs) |
| Primary endpoint | PFI |

# 2. Why This Gene — Not Others

## Primary hypothesis gene(s)

**urine DNA methylation biomarkers:** urine DNA methylation biomarkers (noninvasive tumor shedding), circulating tumor cells (CTCs, systemic dissemination/readout), and immune contexture (operationalized here as TLS-like lymphoid aggregates).

**circulating tumor cells (CTCs):** Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

### Alternative / secondary genes considered

| Tier | Gene | Role / Reason excluded from core |
|---|---|---|
| Secondary | MSH6 | Supporting modifier from knowledge graph |
| Secondary | TGF-β signaling axis | Supporting modifier from knowledge graph |
| Secondary | SDC2 | Supporting modifier from knowledge graph |
| Secondary | TFPI2 | Supporting modifier from knowledge graph |
| Secondary | HDL-C | Supporting modifier from knowledge graph |
| Secondary | Immunoscore (CD3/CD8-based) | Supporting modifier from knowledge graph |

## 3. Clinically Testable Predictions

**Prediction 1:** a local tissue context characterized by TLS-like aggregates and immune composition,

**Prediction 2:** a systemic readout of tumor shedding via noninvasive urine DNA methylation biomarkers and CTCs, and

**Prediction 3:** the tumor intrinsic molecular milieu that can drive invasion and immune evasion. Prior work shows TLS presence can correlate with favorable or unfavorable outcomes depending on context; urine DNA methylation markers (e.g., SDC2/TFPI2 panels) and CTC counts provide orthogonal noninvasive measures of tumor activity and dissemination.

## 4. Minimum Viable Validation Plan

### A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

### B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

### C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

## 5. Strengths, Caveats & Alternative Directions

### Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

### Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

### Alternative Directions

- MSH6: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TGF-β signaling axis: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.